SAT Unit 3.3
One-Variable Data

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Overview

1. Mean, Median, and Mode
2. Standard Deviation and Range
3. Frequency Table, Dot Plot, Bar Graphs, and Histogram

Mean, Median, and Mode

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Understanding Mean, Median, and Mode

Mean (Average):

$$\text{Mean} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

- **Effect of Outliers:** Extremely sensitive to outliers.
- **Interpretation:** Represents a balance point of the data.
- If one value increases by k, the mean increases by $\frac{k}{n}$.

Median (Middle Value):

- The value separating the higher half from the lower half.
- **Robustness:** Resistant to outliers; a better measure of central tendency for skewed data.
- A single extreme value has no effect unless it shifts the order.

More on Mean and Median with Data Behavior

When is Median Preferred over Mean?

- Skewed data (e.g., income distribution).
- Presence of outliers or extreme values.

Examples of Change Impact:

- Add 1000 to an income of $30,000$: Mean jumps; median may stay same.
- Remove a high outlier: Mean drops; median may remain unchanged.

Mode:

- The value(s) that appear most frequently in a data set.
- May have none, one, or multiple modes.
- Useful for categorical or discrete data.

Data Set and Mean Problem

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A:

$$43,\ 45,\ 44,\ 43,\ 38,\ 39,\ 40,\ 46,\ 40$$

The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

Solution to Data Set and Mean Problem

Let the missing number be x. The sum of the 9 known numbers is:

$$43 + 45 + 44 + 43 + 38 + 39 + 40 + 46 + 40 = 378$$

Mean of all 10 numbers must be an integer greater than 42. So:

$$\frac{378 + x}{10} > 42 \Rightarrow 378 + x > 420 \Rightarrow x > 42$$

Since the mean must also be an integer, try smallest integer value for the mean greater than 42, which is 43:

$$\frac{378 + x}{10} = 43 \Rightarrow 378 + x = 430 \Rightarrow x = 52$$

Correct Answer: $\boxed{52}$

Question: Effect of Adding an Outlier to a Data Set

Data set F consists of 55 integers between 170 and 290. Data set G consists of all the integers in data set F as well as the integer 10. Which of the following must be less for data set F than for data set G?

I. The mean

II. The median

A. I only

B. II only

C. I and II

D. Neither I nor II

Answer Explanation: Effect of Adding an Outlier to a Data Set

- Adding an extreme low value (like 10) to data set F will significantly reduce the **mean** of the data set, since the mean is sensitive to outliers.
- However, the **median** is the middle value. Since 10 is far below the range of values in set F (170–290), it will simply be placed at the beginning and not affect the central value of the sorted list.

Conclusion:

- Mean of F $>$ Mean of G
- Median remains unchanged

Correct Answer: $\boxed{A}$

Standard Deviation and Range

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Standard Deviation: Definition and Analysis

Definition:

$$\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2}$$

- Measures the **spread** or **dispersion** of data around the mean.
- A small σ indicates data points are close to the mean.
- A large σ implies data is more spread out.

How It Changes:

- Adding a constant to all values: σ stays the same.
- Multiplying all values by a constant k: σ becomes $|k|\sigma$.
- Introducing an outlier: σ increases significantly.

Range: Definition and Behavior

Definition:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

Properties:

- Measures the **total spread** of a data set.
- Only uses the two **extreme values**; does **not** reflect the distribution of other data points.
- **Sensitive to outliers** — a single large or small value can drastically change the range.

How Range is Affected:

- Adding a constant to all values: **No change** in range.
- Multiplying all values by a positive constant k: Range becomes $k \cdot (\text{original range})$.
- Replacing max or min with a more extreme value increases the range.

Modified Data Set and Statistical Measures

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median.

Which of the following measures does **NOT** have the same value in both the original and new data sets?

A. Median

B. Mean

C. Sum of the numbers

D. Standard deviation

Answer Explanation

- The **median** stays the same, since the middle number (33) is unchanged.
- The **mean** unchanged.
- The **sum** of the values changes for the same reason the mean does — because of the increases and decreases.
- The **standard deviation** increases, since numbers are being moved further from the center (more spread).

The only measure that does **not** stay the same is the standard deviation.
Correct Answer: $\boxed{D}$

Frequency Table, Dot Plot, Bar Graphs, and Histogram

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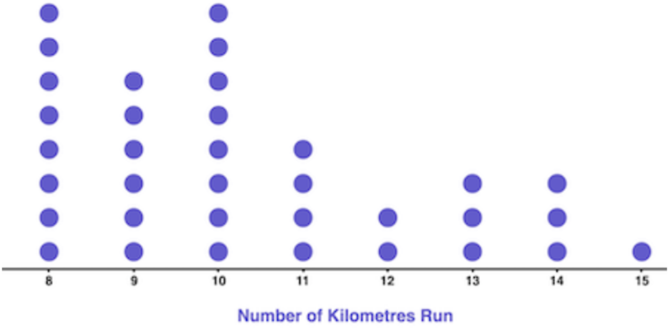
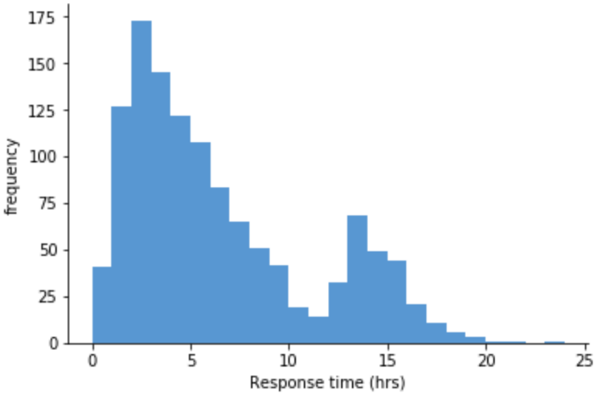
Dot Plot

A **dot plot** is a simple way to display numerical data using dots. Each dot represents one occurrence of a data value, and the dots are stacked vertically above a number line corresponding to the data values.



Histogram

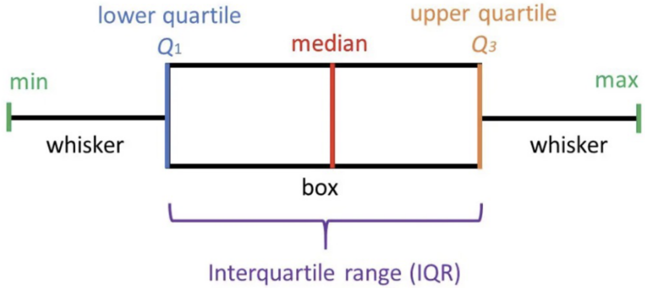
A **Histogram** is a type of bar graph that represents the **frequency distribution** of a dataset. It displays data in intervals (also called bins), where the height of each bar corresponds to the frequency of data within that interval. Unlike bar charts, there are **no gaps** between the bars in a histogram.



Box Plot

A **box plot**, also known as a box-and-whisker plot, graphically represents the **five-number summary**:

Minimum, Q_1 (First Quartile), Median, Q_3 (Third Quartile), Maximur



Contingency Table (Two-Way Table)

A **Contingency Table** or **Two-Way Table** summarizes the frequency of occurrences for two categorical variables. It helps analyze the relationship between these variables by organizing data into rows and columns.

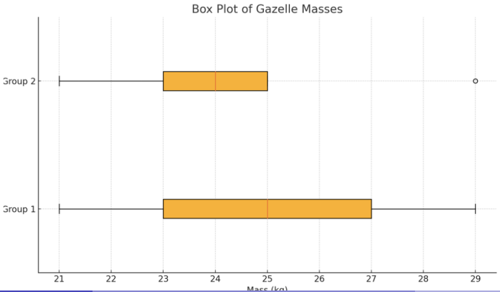
The table below shows the relationship between students' **Gender** and their **Preference for Study Subject**.

Gender	Math	Science	History	Total
Male	20	25	15	60
Female	30	20	40	90
Total	50	45	55	150

Box Plot Interpretation – Mass Comparison

The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements **must be true**?

- **A.** The mean mass of group 1 is greater than the mean mass of group 2.
- **B.** The mean mass of group 1 is less than the mean mass of group 2.
- **C.** The median mass of group 1 is greater than the median mass of group 2.
- **D.** The median mass of group 1 is less than the median mass of group 2.



Answer Explanation

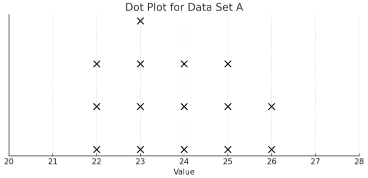
- The median of group 1 is clearly to the right of the median of group 2.
- Since median values are shown as vertical lines inside each box, it is evident that:

$$\text{Median}_{\text{Group 1}} > \text{Median}_{\text{Group 2}}.$$

- Statements about the mean **cannot** be verified from a box plot alone.

Correct Answer: $\boxed{C}$

Dot Plot and Data Transformation



The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A.

Which of the following correctly compares the medians and the ranges of data sets A and B?

A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

D. The median of data set B is greater than the median of data set A

Answer Explanation

- Adding the same number to all values in a data set increases the median by that amount.
- However, the spread (range) remains unchanged because both the maximum and minimum values are increased equally.
- Thus:
  - **Median of B** is greater than **median of A**
  - **Range of B** is equal to **range of A**

Correct Answer: $\boxed{C}$

Median Comparison – Grouped Data

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

Score	Group A	Group B
2	1	0
3	1	0
4	2	0
5	1	4
6	3	2
7	0	0
8	0	2
9	1	1
10	0	2
Total	9	11

Question: The median of the scores for group B is how much greater

Answer – Median Comparison

To find the medians:

- **Group A (9 players):** Median is the 5^{th} value.
  Sorted values based on frequency:

  $$2,\ 3,\ 4,\ 4,\ \boxed{5},\ 6,\ 6,\ 6,\ 9$$

  Median of Group A $= \boxed{5}$

- **Group B (11 players):** Median is the 6^{th} value.
  Sorted values based on frequency:

  $$5,\ 5,\ 5,\ 5,\ 6,\ \boxed{6},\ 8,\ 8,\ 9,\ 10,\ 10$$

  Median of Group B $= \boxed{6}$

Correct Answer: $\boxed{The median of group B is}\boxed{1}$ greater than the median of group A.

Practice

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Question: Median Comparison of Data Sets P and Q

Data set P: 41, 43, 45, 45, 47, 47, 51, 51, 55
Data set Q: 33, 33, 36, 36, 38, 38, 40, 40, 40
The data values for data sets P and Q are shown. The median of data set P is how much greater than the median of data set Q?

Answer Explanation: Median Comparison of Data Sets P and Q

Step 1. Find the median of data set P:
Data set P (ordered): 41, 43, 45, 45, 47, 47, 51, 51, 55
Median: 47
Step 2. Find the median of data set Q:
Data set Q (ordered): 33, 33, 36, 36, 38, 38, 40, 40, 40
Median: 38
Step 3. Difference:
$47 - 38 = 9$
Correct Answer: $\boxed{9}$

Question: The frequency table summarizes the prices of 30 items...

The frequency table summarizes the prices of 30 items in a shopping cart.

Price (dollars)	Frequency
25	8
40	2
45	4
55	10
70	6

If an item with a price of 25 dollars is removed from the shopping cart, what will be the effect on the mean and median price of the items in the shopping cart?

- **A.** The mean and median will both decrease.
- **B.** The mean and median will both increase.
- **C.** The mean will decrease, and the median will remain the same.
- **D.** The mean will increase, and the median will remain the same.

Answer Explanation: The frequency table summarizes the prices of 30 items. . .

Step 1. Mean Analysis: Removing a low-priced item (25) from the dataset will increase the overall mean, because the total sum decreases by less than proportionally compared to the decrease in count.
Step 2. Median Analysis: Since there are 30 items initially, the median is the average of the 15th and 16th values when ordered. After removing one item priced at 25 (the lowest), there are now 29 items. The new median is the 15th value, which remains at 55.
Conclusion: The mean will increase, and the median will remain the same.
Correct Answer: $\boxed{D}$

Question: Bus Travel Time Comparison

The tables give the distribution of daily travel times between two towns for bus A and bus B over a 40-day period.

Bus A		**Bus B**	
Travel time (minutes)	Frequency	Travel time (minutes)	Frequency
44	5	25	5
45	10	30	10
47	15	35	15
48	10	40	10

Which statement correctly compares the standard deviations of travel times for bus A and bus B?

A. The standard deviation of travel times for bus A is less than the standard deviation of travel times for bus B.

B. The standard deviation of travel times for bus B is less than the standard deviation of travel times for bus A.

C. The standard deviation of travel times for bus A is equal to the standard deviation of travel times for bus B.

Answer Explanation: The tables give the distribution of daily travel times...

Step 1. Check the range of values:

- Bus A: 44 to 48 (Range = 4)
- Bus B: 25 to 40 (Range = 15)

Step 2. Analyze spread:

- Bus A values are clustered closer together.
- Bus B values are more spread out.

Conclusion: Bus B has a larger standard deviation than Bus A.
Correct Answer: $\boxed{A}$

Question: Comparing Frequency Tables for Data Set A

Data set A

Value	Frequency
30	11
50	11
70	11

The frequency table shown represents data set A. Which of the following frequency tables represents a data set that has a mean greater than the mean of data set A?

A.

Value	Frequency
30	12
50	12
70	12

B.

Value	Frequency
30	9
50	10

Answer Explanation: Comparing Frequency Tables for Data Set A

Step 1. Mean of data set A:

$$\text{Mean} = \frac{(30 \cdot 11) + (50 \cdot 11) + (70 \cdot 11)}{33} = \frac{330 + 550 + 770}{33} = \frac{1650}{33} = 50.$$

Step 2. Compare each option:

- A: All frequencies are equal, mean $= 50$ (same as set A).
- B: More weight on 70 and less on 30, so mean ¿ 50.
- C: More weight on 30, mean ¡ 50.
- D: Equal frequencies on 30 and 70 but same as set A on 50; mean $=$ 50.

Correct Answer: $\boxed{B}$

Question: Comparing Data Sets A and B

Data set A		Data set B	
Value	**Frequency**	**Value**	**Frequency**
10	2	13	2
11	3	14	3
12	7	15	7
13	3	16	3
14	9	17	9
15	5	18	5

The tables show the frequencies of the values for data sets A and B. Which of the following statements best compares the means and the standard deviations of data sets A and B?

A. The mean of data set B is equal to the mean of data set A, and the standard deviation of data set B is equal to the standard deviation of data set A.

B. The mean of data set B is equal to the mean of data set A, and the standard deviation of data set B is greater than the standard

Answer Explanation: Comparing Data Sets A and B

Step 1. Compare the means:

- Data set B is a translation of data set A by adding 3 to each value.
- Therefore, the mean of data set B is 3 more than the mean of data set A.

Step 2. Compare the standard deviations:

- Adding a constant shifts all data points equally, so standard deviation does not change.
- Therefore, the standard deviation of data set B is equal to that of data set A.

Conclusion:

- The mean of data set B is greater than the mean of data set A.
- The standard deviations of data sets A and B are equal.

Correct Answer: $\boxed{C}$

Thank You!

We appreciate your time and focus.

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